MATH 152
Mrs. Bonny Tighe

EXAM I

6.1-7.5
50 points
PART TWO

NAME Solution

TA_____ Wed 2/28/18

1. (10 points) Integrate $\int \cancel{\tan^{-1} x\,dx}\ \sin\sqrt{x}\,dx$

$m=\sqrt{x}$

$dm = \frac{1}{2\sqrt{x}}\,dx$

$2\sqrt{x}\,dm = dx$

$2m\,dm = dx$

$\int \sin m\ 2m\,dm$

$u=2m$	$dv=\sin m\,dm$
$du=2dm$	$v=-\cos m$

$-2m\cos m - \int -\cos m\ 2dm$

$-2m\cos m + 2\sin m + C$

$-2\sqrt{x}\cos\sqrt{x} + 2\sin\sqrt{x} + C$

2. (10 points) Integrate: $\int_{\sqrt{3}}^{2} \frac{\sqrt{4-x^2}}{x^2}\,dx$

$x = 2\sin\theta$

$dx = 2\cos\theta\,d\theta$

$2 = 2\sin\theta \quad \theta = \pi/2$

$\sqrt{3} = 2\sin\theta \quad \theta = \pi/3$

$\int_{\pi/3}^{\pi/2} \frac{\sqrt{4-4\sin^2\theta}}{4\sin^2\theta}\ 2\cos\theta\,d\theta$

$\int_{\pi/3}^{\pi/2} \frac{\sqrt{4(1-\sin^2\theta)}}{4\sin^2\theta}\ 2\cos\theta\,d\theta = \int_{\pi/3}^{\pi/2} \frac{2\cos\theta\ 2\cos\theta\,d\theta}{4\sin^2\theta} =$

$\int_{\pi/3}^{\pi/2} \cot^2\theta\,d\theta = \int_{\pi/3}^{\pi/2} (\csc^2\theta - 1)\,d\theta = -\cot\theta - \theta \Big]_{\pi/3}^{\pi/2}$

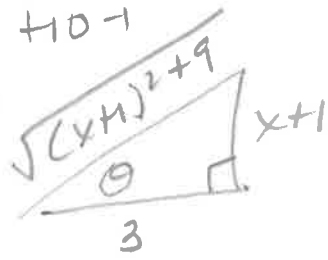
$(-\cot\pi/2 - \pi/2) - (-\cot\pi/3 - \pi/3)$

$0 - \pi/2 + \sqrt{3}/3 + \pi/3 = \sqrt{3}/3 - \pi/6$

3. (10 points) Integrate: $\int \frac{dx}{\sqrt{x^2+2x+10}}$

$x^2+2x+1+10-1$

$(x+1)^2+9$



$$\int \frac{dx}{\sqrt{(x+1)^2+9}}$$

$(x+1) = 3\tan\theta$

$dx = 3\sec^2\theta\, d\theta$

$$\int \frac{3\sec^2\theta\, d\theta}{\sqrt{9\tan^2\theta+9}} = \int \frac{3\sec^2\theta\, d\theta}{\sqrt{9(\tan^2\theta+1)}} = \int \frac{3\sec^2\theta\, d\theta}{3\sec\theta}$$

$$\int \sec\theta\, d\theta = \ln|\sec\theta + \tan\theta| + C$$

$$\ln\left|\frac{\sqrt{(x+1)^2+9}}{3} + \frac{x+1}{3}\right| + C$$

4. (10 points) Integrate $\int \frac{x+3}{x^4+x^2}\, dx$

$$\int \frac{x+3}{x^2(x^2+1)}\, dx \qquad \frac{A}{x} + \frac{B}{x^2} + \frac{Cx+D}{x^2+1} = \frac{x+3}{x^2(x^2+1)}$$

$$Ax(x^2+1) + B(x^2+1) + x^2(Cx+D) = x+3$$

$$Ax^3 + Ax + Bx^2 + B + Cx^3 + Dx^2 = x+3$$

$$x^3(A+C) + x^2(B+D) + x(A) + B = x+3$$

$C = -1 \qquad D = -3 \qquad A = 1 \qquad B = 3$

$$\int \frac{1}{x} + \frac{3}{x^2} + \frac{-x-3}{x^2+1}\, dx$$

$$\int \left(\frac{1}{x} + 3x^{-2} - \frac{x}{x^2+1} - \frac{3}{x^2+1}\right) dx$$

$$\ln|x| - \frac{3}{x} - \frac{1}{2}\ln|x^2+1| - 3\tan^{-1}x + C$$

5. (10 points) Integrate $\cancel{\int \sin\sqrt{x}\,dx}$ $\int x \sin^2 x\,dx$

$u = x \quad dv = \left(\frac{1}{2} - \frac{1}{2}\cos 2x\right)dx$

$du = dx \quad v = \frac{1}{2}x - \frac{1}{4}\sin 2x$

$x\left(\frac{1}{2}x - \frac{1}{4}\sin 2x\right) - \int\left(\frac{1}{2}x - \frac{1}{4}\sin 2x\right)dx$

$\frac{1}{2}x^2 - \frac{x}{4}\sin 2x - \left[\frac{1}{4}x^2 + \frac{1}{8}\cos 2x\right] + C$

$\frac{1}{2}x^2 - \frac{x}{4}\sin 2x - \frac{1}{4}x^2 - \frac{1}{8}\cos 2x + C$

$\frac{1}{4}x^2 - \frac{1}{4}x\sin 2x - \frac{1}{8}\cos 2x + C$

6. (5 points) True or False.

a) To evaluate $\int \sin^2(w)dw$ you need to use u-substitution. False

b) An average value can be negative. True

c) $\int x^2 \cos(x^3)dx$ has to be evaluated using parts. False

d) The partial fraction decomposition, without finding the constants, for $\frac{1}{x^2(x+1)}$ would be $\frac{Ax+B}{x^2} + \frac{C}{x+1}$ True

e) When trying to find the area between $y = \sin x$ and $y = \sin 2x$ on the interval $[0, \frac{\pi}{4}]$ you would need to set up two separate integrals. False

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EXAM IA

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PART TWO

NAME Solutions

TA_____ Wed 2/28/18

1. (10 points) Integrate $\int \frac{1}{\sqrt{x^2+2x+5}}\,dx$

$x^2+2x+1+5-1$

$(x+1)^2+4$

$(x+1) = 2\tan\theta$

$dx = 2\sec^2\theta\, d\theta$

$$\int \frac{1}{\sqrt{(x+1)^2+4}}\,dx$$

$$\int \frac{1}{\sqrt{4\tan^2\theta+4}}(2\sec^2\theta\, d\theta) = \int \frac{2\sec^2\theta\, d\theta}{\sqrt{4(\tan^2\theta+1)}} = \int \frac{2\sec^2\theta\, d\theta}{2\sec\theta}$$

$$\int \sec\theta\, d\theta = \ln|\sec\theta + \tan\theta| + C$$

$$\ln\left|\frac{\sqrt{(x+1)^2+4}}{2} + \frac{(x+1)}{2}\right| + C$$

2. (10 points) Integrate: $\int t\cos^2 t\, dt$

$u = t \qquad dv = \frac{1}{2} + \frac{1}{2}\cos 2t\, dt$

$du = dt \qquad v = \frac{1}{2}t + \frac{1}{4}\sin 2t$

$$t\left(\tfrac{1}{2}t + \tfrac{1}{4}\sin 2t\right) - \int\left(\tfrac{1}{2}t + \tfrac{1}{4}\sin 2t\right)dt$$

$$\tfrac{1}{2}t^2 + \tfrac{t}{4}\sin 2t - \left[\tfrac{1}{4}t^2 - \tfrac{1}{8}\cos 2t\right] + C$$

$$\tfrac{1}{2}t^2 + \tfrac{1}{4}t\sin 2t - \tfrac{1}{4}t^2 + \tfrac{1}{8}\cos 2t + C$$

$$\tfrac{1}{4}t^2 + \tfrac{1}{4}t\sin 2t + \tfrac{1}{8}\cos 2t + C$$

3. (10 points) Integrate: $\int_{\sqrt{3}}^{2} \frac{\sqrt{4-x^2}}{x^2}\,dx$

$x = 2\sin\theta$ $\quad 2 = 2\sin\theta \quad \theta = \pi/2$

$dx = 2\cos\theta\, d\theta$ $\quad \sqrt{3} = 2\sin\theta \quad \theta = \pi/3$

$$\int_{\pi/3}^{\pi/2} \frac{\sqrt{4-4\sin^2\theta}}{4\sin^2\theta}\, 2\cos\theta\, d\theta = \int_{\pi/3}^{\pi/2} \frac{\sqrt{4(1-\sin^2\theta)}}{4\sin^2\theta}\, 2\cos\theta\, d\theta$$

$$\int_{\pi/3}^{\pi/2} \frac{2\cos\theta\, 2\cos\theta\, d\theta}{4\sin^2\theta} = \int_{\pi/3}^{\pi/2} \cot^2\theta\, d\theta = \int_{\pi/3}^{\pi/2} (\csc^2\theta - 1)\, d\theta$$

$$-\cot\theta - \theta \Big]_{\pi/3}^{\pi/2} = (-\cot\pi/2 - \pi/2) - (-\cot\pi/3 - \pi/3) =$$

$$(-0 - \pi/2) - (-\sqrt{3}/3 - \pi/3) = \sqrt{3}/3 - \pi/6$$

4. (10 points) Integrate $\int \frac{x+2}{x^4+x^2}\,dx$

$$\int \frac{x+2}{x^2(x^2+1)}\,dx = \int \left(\frac{A}{x} + \frac{2}{x^2} + \frac{-x-2}{x^2+1} \right) dx$$

$$\frac{A}{x} + \frac{B}{x^2} + \frac{Cx+D}{x^2+1} = \frac{x+2}{x^2(x^2+1)}$$

$$Ax(x^2+1) + B(x^2+1) + x^2(Cx+D) = x+2$$

$$Ax^3 + Ax + Bx^2 + B + Cx^3 + Dx^2 = x+2$$

$$x^3(A+C) + x^2(B+D) + Ax + B = x+2$$

$C = -1 \quad D = -2 \quad A = 1 \quad B = 2$

$$\int \left(\frac{1}{x} + 2x^{-2} - \frac{x}{x^2+1} - \frac{2}{x^2+1} \right) dx$$

$$\ln|x| - \frac{2}{x} - \frac{1}{2}\ln|x^2+1| - 2\tan^{-1}x + C$$

5. (10 points) Integrate $\int e^{\sqrt{x}}\,dx$

$m = \sqrt{x}$

$dm = \frac{1}{2\sqrt{x}}\,dx$

$2\sqrt{x}\,dm = dx$

$2m\,dm = dx$

$\int e^{m}\,2m\,dm$

$u = 2m \qquad dv = e^{m}dm$

$du = 2dm \qquad v = e^{m}$

$2me^{m} - \int e^{m}(2dm) = 2me^{m} - 2e^{m} + C$

$2\sqrt{x}\,e^{\sqrt{x}} - 2e^{\sqrt{x}} + C$

6. (5 points) True or False.

a) To evaluate $\int \cos^3(w)\,dw$ you need to use u-substitution. True

b) An average value can only be a positive value . False

c) $\int x^2 e^{x^3}\,dx$ has to be evaluated using parts. False

d) The partial fraction decomposition, without finding the constants, for $\frac{1}{x^2(x+1)}$ would be $\frac{Ax+B}{x^2} + \frac{C}{x+1}$ True

e) When trying to find the area between $y = \sin x$ and $y = \sin 2x$ on the interval $[0, \pi]$ you would need to set up two separate integrals. True